

HAT-P-37b

R-2632

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_200713 · created 2026-08-29T00:33:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459043.8232 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.05
Transit depth (Rp/Rs)^2	2.24 +/- 1.48 [%]
Orbital Inclination (inc)	89.01 +/- 2.41
Airmass coefficient 1 (a1)	0.076 +/- 0.046
Airmass coefficient 2 (a2)	2.19 +/- 1.08
Scatter in the residuals of the lightcurve fit is	54.93 %
Best Comparison Star	#1 - [538, 304]
Optimal Aperture	3.02
Optimal Annulus	9.4
Transit Duration (day)	0.099 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.15 $\pm$ 0.05 vs 0.1394 (deviation 0.21 σ)
Duration fit vs published	0.099 d vs 0.0983 d (deviation 0.06 σ)
Mid-time O-C	-14.4 min
Depth SNR	3.0
Residual scatter	54.93 %

Light curve

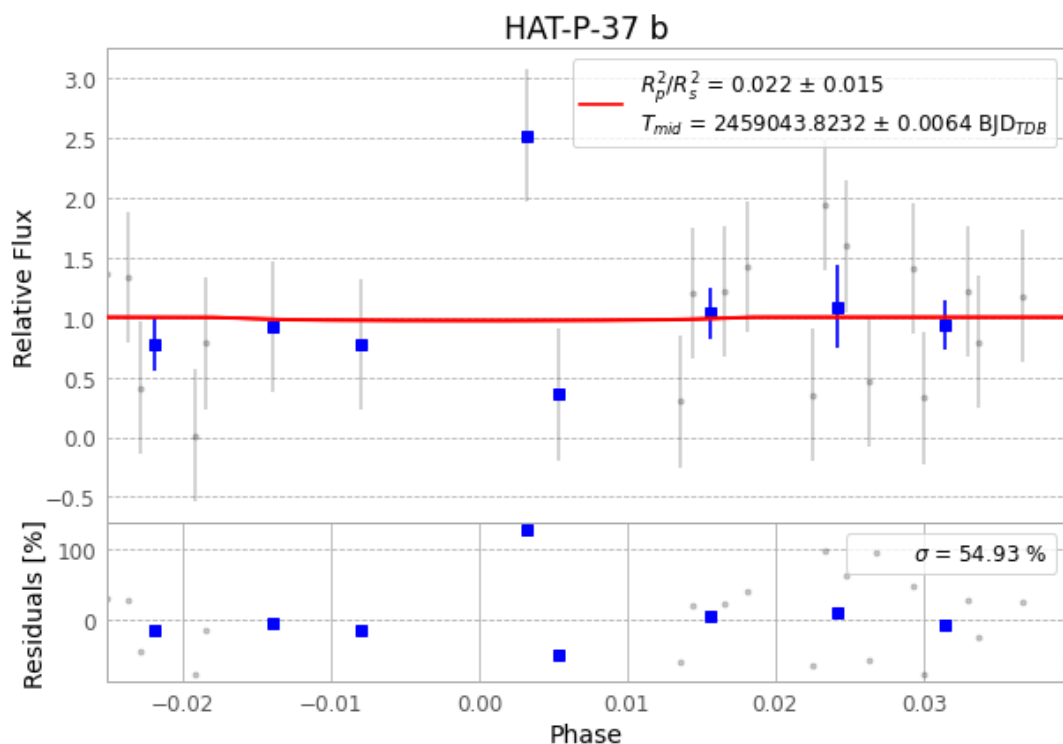


Figure 1 — FinalLightCurve_HAT-P-37 b_2020-07-12.png (SHA-256 b0cff63223967671...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 145], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 af7b3255d2b17afc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

88 FITS frames (58 MB), HATP-37200713060012.FITS → HATP-37200713102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-200713.log (ecb1eef7520e...), FinalLightCurve_HAT-P-37 b_2020-07-12.png (b0cff6322396...), FinalLightCurve_HAT-P-37 b_2020-07-12.csv (b7c31c6b64d7...)

Compute resources

Wall time 138.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:33Z.